Data Processing in Python (Fall 2026)

Workshop Leads: Maria Haddad and Sam Nittala

Registration link

Approximate duration: 2 hours

Prerequisites:

- Basic knowledge of Python is required.
- Attendees must be comfortable using variables for simple data types, as well as collections. Attendees should also be comfortable with loops and control flow and be familiar with the basics of using functions in Python.
- To be able to participate in the exercises, participants must either:
    - Have a Google Account to run the material in-browser as a Colab notebook **(preferred)**
    - Have a local installation of Python and a tool to edit and run Jupyter notebooks (e.g., Jupyter Lab, Microsoft Visual Studio Code or PyCharm)

Summary: (2-3 sentences summarizing the workshop)

In this 2-hour workshop, students will learn basic data processing skills using Python. Attendees will learn how to import code from other modules and packages to take advantage of the existing Python ecosystem. After seeing how to access packages, we will explore popular data analysis packages. We will see how to use NumPy to perform operations on large data arrays and how to use Matplotlib to generate clear data visualizations. We will also scratch the surface on using pandas to store data in tables. Along the way, we will discuss how to approach new, unfamiliar packages and learn how to use them.

Learning Objectives: (List 3-5 learning objectives participants will learn upon completion of this workshop)

1. Import code from existing modules and packages.
2. Use NumPy to easily process multidimensional data.
3. Use Matplotlib to generate different types of plots to visualise data.
4. Use pandas to represent data stored in tables.
5. Approach a new package and explore its documentation and examples.

Content:

1. **Module 1 – Modules and Packages (20 minutes)**
    a. Using Modules
        i. What is a module?
        ii. Importing a module
        iii. Importing specific functions
    b. A Brief Intro to Package Management
        i. What is a package?
        ii. Installing packages using conda
        iii. Installing packages using pip
        iv. (Optional) Jupyter notebook trick – Magic commands
        v. (Optional) Other installation tips
        vi. Importing packages
    c. **Exercises**
2. **Module 2 – Introduction to NumPy Arrays (30 minutes)**
    a. Introducing NumPy
        i. What is an array?
        ii. NumPy fundamentals
        iii. Creating and reshaping arrays
        iv. Indexing arrays
        v. Simple assignments
    b. Array Operations
        i. Element-wise arithmetic
        ii. Statistics on arrays
        iii. Combining arrays
        iv. Exploring the documentation
    c. **Exercises**
3. **Module 3 – Visualizing Data with Matplotlib (30 minutes)**
    a. Creating Plots with Matplotlib
        i. Introduction to pyplot
        ii. Creating simple plots, bar plots, histograms, …
        iii. Customizing plots: Labelling axes, titles, and more!
        iv. Advanced plots: scatter plots, image plots
        v. Exporting plots as images

vi. Generating subplots

b. Exploring the Matplotlib Documentation

c. **Exercises**

4. **Module 4 – Intro to Tabular Data with Pandas (30 minutes)**
   a. Fundamentals of pandas
      i. Motivation: When arrays aren't enough…
      ii. Introducing the Series and DataFrame
      iii. Intro to Grouping
      iv. Loading and saving tables
   b. Exploring the pandas Documentation
   c. **Exercise**
5. **Module 5 – A Brief Guide to Exploring the Unknown (10 minutes)**
   a. What to learn next? How?
   b. How to get help and how not to get help
      i. Your code editor
      ii. Documentation
      iii. Books
      iv. Tutorials
      v. Stack Overflow (and pitfalls)
      vi. ChatGPT (and pitfalls)
   c. Other cool programming